Statement of Work

Cooling Tower and Support Replacement

1.0 Background

1.1.1 The Department of Treasury's Bureau of Engraving and Printing (BEP), Washington D.C., Facility (DCF) is comprised of the Main Building and the Annex Building. The mission of the BEP is to manufacture and produce United States currency.

The Annex Building was erected in 1937 and features a limestone façade, constructed entirely of reinforced concrete. The Annex Building features a central corridor with five wings that span from north to south along this line. The building is between 13th and 14th Streets and C and D Streets in the southwest quadrant of Washington, DC.

The Main Building, constructed in 1914, features a neoclassical design with an exterior of Indiana Limestone and granite. It is located between 14th and 15th Streets in the Southwest quadrant of Washington, DC.

The existing structural steel platform has reached a corroded and rusted condition and needs to be replaced with a new structural steel platform. The roof of the chiller / power plant wing located between A-Wing and B-Wing of the Main Building, is deteriorating and has cracks and leaks are present in multiple areas. As a result of this aging, water intrusion through the roofing layers into the building interiors impacts the functionality of the spaces below and causes damage and hazardous conditions. This roof replacement shall be demolished and replaced as part of the Cooling Towers replacement project, and is expected to be performed between the demolishing of the existing steel platform and the construction of the new steel platform

This Statement of Work (SOW) defines the scope, requirements, and deliverables, including, but not limited to, a complete assessment of the scope of work, recommendation of the scope of work, demolition, and the replacement of four (4) two cell existing Evapco cooling towers (Model: AT 424-936)**(BRAND NAME OR EQUAL)**, and replacement of one (1) two cell BAC cooling tower (model number is VXT-800CR **(BRAND NAME OR EQUAL)** from Baltimore air coil company, Serial number is 84-5979) associated structural steel platform, roofing, associated structural components, associated mechanical components, associated electrical components, pump, motors, piping, chemical feed, fire/smoke detector system, condenser water motor/pumps, electrical circuit breaker, piping insulation, SCADA integration, chillers, and associated control systems. The new cooling towers shall provide capacity to support the four chillers able to handle the four (4) chillers at maximum load, as each chiller is rated at 1,375 tons. The contractor shall provide recommendations for upgrading each chiller and its tonnage, including pumps, motors, piping, coils, and other associated components, to accommodate the newly installed cooling tower.

The new cooling tower for the BAC replacement should be able to handle one chiller at a maximum load of 1,375 tons or more during the slow season, when only one chiller is operating. This cooling tower should be capable of providing free cooling. Each cooling tower should not exceed the current cooling tower's maximum operating weight. Contractor shall ensure that the project adheres to all guidelines and code, including but not limited to the International Plumbing Code (IPC), the American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE) standard and guideline, the American Society of Mechanical Engineers (ASME) codes and standards, and the National Electrical Code, etc.

The contractor shall provide As-built drawings, including updated architectural, structural, and Mechanical, Electrical and Plumbing (MEP) plans upon completion of construction and updated schematics for this project. The contractor shall provide an integration and SCADA commissioning, demolition, and disposal report. The contractor should provide equipment, structural, and all necessary submittals and cut sheets for all replacement components. The work shall be performed by a qualified HVAC/mechanical contractor with demonstrated experience in large-scale condenser water system installations, including the integration of cooling towers with SCADA systems.

The contractor shall survey the chiller plant, including, but not limited to, all associated chiller components and accessories, piping, pumps, motors, AHUs, MAUs, FCUs, etc. The contractor should provide a recommendation on what type of chillers is best suited for our mission. This recommendation includes, but is not limited to, brand, tonnage capacity, footprint, operating weight, efficiency metrics, maintenance costs, compressor performance, and operational costs. This recommendation shall be based on the newly installed cooling tower.

2.0 Purpose

2.1 **Product, Service or Outcome Needed.**

The objective is to obtain construction services for the removal, disposal, and replacement of the cooling towers associated structural steel platform, mechanical, pump, motors, piping, chemical feed, fire, and control systems located in the BEP main building. In addition, replace/repair the Power Plant Wing Roof leaking issues to eliminate potential damages, safety hazard, and possible BEP's operation disruptions to the spaces below this roof.

2.2 **Scope of the Product, Service, or Outcome.**

This project is a Design–Build. The contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform a survey and provide approved drawings for the existing Power plant wing structural steel platform and then the demolition for this structural steel platform, mechanical, electrical, integration, Power Plant Wing roof replacement, and HVAC Replacements as defined in this Statement of Work, except for those items specified as government-furnished property and services. The contractor shall submit a proposal that includes the total cost and price break down for each item to

this statement of work (SOW) as delineated, along with all labor, materials, equipment, tools, supplies, supervision, and general construction services to design and install as noted in the SOW.

Non-Personal Service statement: The Contractor employees performing services under this Contract will be controlled, directed, and supervised at all times by management personnel of the Contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in the Contract. Contractor employees will perform their duties independently of, and without the supervision of, any Government official or other Government Contractor. The tasks, duties, and responsibilities outlined in this Contract may not be interpreted or implemented in any manner that results in any Contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions outlined in Parts 7.5 and 37.1 of the FAR.

In accordance with FAR 36.204, Disclosure of the Magnitude of Construction Projects, advance notice and solicitations shall state the magnitude of the requirement in terms of physical characteristics and estimated price range. The estimated price range is between $1,000,000 and $5,000,000.

3.0 Technical Requirements/Tasks/Outcomes

3.1 **Basic Services.**

The contractor shall provide services for all phases of the project, and each phase shall be subject to BEP approval at designated project milestones. The BEP shall remain fully operational during all phases of construction. This project includes (but is not limited to) all labor, materials, supplies, equipment, services, training, communications, transportations, shipping, supporting steel structure, HVAC controls, operation and maintenance manuals, as-built drawings, construction, testing, adjusting, balancing, and commissioning. In conjunction with the aforementioned, the Contractor shall perform the following as further detailed in the Statement of Work (SOW). Plans are not available for areas in which work is to be performed, but the BEP will try to provide, but cannot guarantee, archive drawings for "Informational Purposes Only". It is the Contractor's responsibility to gather all information necessary for the successful completion of this project, including but not limited to a thorough understanding of the existing conditions under which the work will be performed. Before commencing the work, the Contractor shall conduct a design.

The design shall provide set of independent work package designs that together specify how to fulfill the intent and purpose of this contract. The work packages will be identified and selected in a way that allows fast tracking without risk to other work packages or the overall design. Hereafter, the unqualified word, "design," refers to either the overall design or an independent work package design depending on context.

3.1.1.1 Contractor shall thoroughly survey the project site and all existing conditions.

3.1.1.2 Contrator shall become familiar with the project layout, conditions, equipment, components, and other attributes having or likely to have an impact on the project.

3.1.1.3 Contractor shall verify existing capacities, make recommendations and provide submittals of proposed units inclusive of proposed capacities.

3.1.1.4 Contractor shall be responsible for Lead Based Paint (LBP) identification, testing/analysis and all associated LBP abatement procedures and requirements. If LBP is located, the Contractor shall note the LBP location on the project drawings, draft a report detailing the location(s) of the LBP and draft a detailed LBP Abatement Plan. The LBP Abatement plan shall be forwarded to the Office of Environment, Health & Safety (OEHS) for review and approval at least two (2) weeks prior to beginning any abatement procedures. No LBP abatements will proceed w/o written approval from the OEHS. The removal of the waste LBP shall be performed by the Contractor IAW EPA and DC Hazardous Waste Disposal regulations and per Section 0113000 - Removal and Disposal of Lead-Based Paint.

3.2 **Survey / Demolition Work.**

3.2.1 **Mechanical/ Electrical/ Controls-**

3.2.1.1 Demolition Plan (Drawings and Specifications): Contractor shall provide detailed plans indicating specific on the mechanical/ Electrical/ Controls and all other associated items. A comprehensive timeline, broken down by specific areas (e.g., Main Gable, Awning Area, etc.), outlining start and completion dates for each demolition phase and key milestones.

3.2.1.2 The Contractor shall coordinate all utility disconnections and reconnections (mechanical, electrical, plumbing, controls) required.

3.2.1.3 Demolish four (4) eight cell existing Evapco cooling towers (Model: AT 424-936), one (1) two cell Baltimore Air Coil cooling tower (model number is VXT-800CR from BAC, Serial number is 84-5979), including but not limited to associated mechanical, support structural, controls, and electrical components.

3.2.1.4 Demolish existing condenser water pumps (Pumps; 75 HP, 230/460V, 1780 RPM) and (Pump 100 HP, 230/460V, 1780), but not limited to any associated existing piping and electrical infrastructure, etc.

3.2.1.5 Demolish the existing condenser water motor/pump (1B, 2B, 3B, 4B), including but not limited to electrical components, piping, motors, insulation, VFDs, controls, etc.

3.2.1.6 Demolish the existing condenser chemical feed system and all associated components.

3.2.1.7 Demolish and remove the abandoned cooling tower labeled as Tower #5. Demo all associated items, including but not limited to piping, electrical components, and

support structures, shall also be removed, and the area restored to a safe and code-compliant condition

3.2.1.8 De-integrate and demolish all outdated controls associated with the existing cooling towers, valves, pumps, etc.

3.2.1.9 Demolish any electrical components associated with this project, including but not limited to circuit breakers, wiring, capacitors, relays, fuses, electrical panel, control valve, sensors, low voltage components, etc.

3.3.1.10 Demolish all existing controls that are associated with the condenser water pump, cooling tower, piping, fire alarm system, Scada system, VFD, raceway, including but not limited to valves, actuators, control panels, switches, sensors, fire alarm smoke system, relays, contactors, limit switches, leak detectors, etc.

3.2.2 **Structural/ Civil.**

3.2.2.1 Survey existing structural steel platform and prepare an approved set of drawings for all the existing steel structural members, such as, but not limited to, girders, beams, braces, columns, stub-ups, connections, grating, ladders, stairs, handrails, guardrails, ladder cages, and all anchor points for both structural components and for fall protection purposes, etc.

3.2.2.2 Provide and submit a signed and sealed drawings and calculations to include all the survey outcome indicated in 3.3.2.1 along with all related layouts, dimensions, plans, elevations, structural details, and recommended protective and anti-corrosion prime and paint that matches the existing to BEP for review. If existing structural members and cross-sections are different from recent structural steel cross - sections provided on the latest AISC Manual, the new drawings shall include equivalent and updated steel structure cross – sections data that have equivalent or larger capacities relative to the existing structure.

3.2.2.3 Demolish remove, salvage and dispose existing structural steel platform as described in this section (3.3.2.1) in accordance with all the governed environmental and local regulations and standards.

3.2.2.4 **Archectural Accessibility and Life-Safety Compliance Upgrades.**

a. **Objective:**
The contractor shall perform comprehensive compliance evaluations, designs, and structural upgrades for all personnel access and egress routes serving the cooling tower work zone. The objective is to bring all existing, altered, and newly installed architectural access elements into full compliance with modern federal safety and accessibility standards.

b. **Specific Systems and Components Included:**
The contractor's scope of work shall include, but is not limited to, the engineering, procurement, demolition, and installation required to upgrade or entirely rebuild the following components:

- **Facility Access Stairs and Ramps:** Structural remediation or replacement of existing structural stairways leading to the cooling tower platforms to ensure compliant rise, run, tread, and load capacities.
- **Roof Access Ladders:** Replacement or modification of fixed roof ladders, including the integration of compliant cages, landing platforms, or ladder safety systems where required.
- **Guards and Handrails:** Installation or replacement of all perimeter guardrails, intermediate rails, and handrails along access paths and equipment platforms.
- **Egress Doors and Hardware:** Upgrading or replacing affected exit doors, frames, and panic hardware serving the immediate roof or mechanical area to ensure unhindered egress.

c. **Applicable Codes and Reference Standards:**
All work performed under this section must strictly adhere to the latest editions of the following federal guidelines and mandates:
1. **Architectural Barriers Act (ABA)** Accessibility Standards for Federal Facilities.
2. **Occupational Safety and Health Administration (OSHA)** Standards (specifically 29 CFR 1910 Subpart D - Walking-Working Surfaces).
3. **National Fire Protection Association (NFPA) 101** Life Safety Code.
4. **International Building Code (IBC)**, as adopted by the local jurisdiction/agency.

3.3 **New Work.**

3.3.1 **Mechanical/ Electrical-**

3.3.1.1 The capacities of all newly installed mechanical, electrical, controls, fire, and piping systems shall, at a minimum, match those of existing demolished units. Prior to commencing work, the contractor shall verify the existing requirements and submit submittals of proposed units, including proposed capacities.

3.3.1.2 Newly installed interior/exterior piping shall match existing demolished piping dimensions.

3.3.1.3 The Contractor shall provide temporary AC Units during the duration of this project before the start of work, as needed to satisfy the building current load.

3.3.1.4 The Contractor shall select equipment that meets Energy Star/FEMP requirements as mandated by Federal Laws (where applicable). BEP requires

equipment to adhere to new guidelines.

3.3.1.5 The Contractor shall retain the services of a licensed electrical contractor to route wiring and other electrical connections for demolition and installation. The Electrical Contractor shall reuse existing wiring to the maximum extent possible and modify electrical services where necessary. All work shall adhere to the equipment manufacturers' wiring diagrams and the National Electric Code (NEC).

3.3.1.6 The Electrical Contractor shall furnish and install electrical supply, panels, breakers, etc., where required.

3.3.1.7 The Contractor is responsible for any required permitting for cranes, hoists, or lifts, etc.

3.3.1.8 Conduct a comprehensive assessment of the existing mechanical, electrical, and control systems before commencing work.

3.3.1.9 Provide a detailed recommendation report, including specifications for replacement equipment and systems.

3.3.1.10 Obtain all necessary permits and coordinate inspections with local authorities having jurisdiction.

3.3.1.11 Ensure all new systems comply with ASHRAE, NFPA, NEC, and relevant local building codes. All new fire device installation work coordinated with fire-alarm/fire-suppression service contractor.

3.3.1.12 Coordinate all shutdowns and equipment transitions to avoid disruption to building operations.

3.3.1.13 Provide complete documentation, testing, commissioning, and training for all installed systems.

3.3.1.14 The cooling tower manufacturer shall provide all specified options and accessories, including but not limited to access ladders, access platforms, cold water basin, and interior walkways, with **stainless steel construction** throughout, in full compliance with guidelines within the Whole Building Design Guide (WBDG) for federal facilities.

3.3.1.15 Furnish and install new cooling towers of the same model or approved equivalent (Model: AT 424-936) meeting all manufacturer specifications, structural support, site performance requirements, and meet the demand of the current chillers (4) rated each at 1375 tons at the plant. **The selected model of cooling tower shall be equipped with high-efficiency drift eliminators that limit drift loss to less than 0.002% of the recirculating water flow.**

3.3.1.16 Furnish and install one (1) two-cell Baltimore Air Coil cooling tower or something compatible that can handle the 1375 tons, with the same weight capacity for operating and stand-alone no operating, and height (current model number is VXT-800CR from BAC, Serial number is 84-5979). This cooling tower shall meet all manufacturer specifications, structural support, and site performance requirements. This cooling tower should be able to handle a load demand of 1,375 tons. **The selected model of cooling tower shall be equipped with high efficiency drift eliminators that limit drift loss to less than 0.002% of the recirculating water flow rate.**

3.3.1.17 All newly installed cooling tower performance should be de-rated due to the towers being surrounded on three (3) sides by the building

3.3.1.18 Install and calibrate ultrasonic flow meters on cooling tower makeup and blowdown lines, integrate conductivity sensors with automated blowdown valve control, and provide data logging for 24/7 monitoring, ensuring all components meet DC Water standards for accurate water management and conservation compliance

3.3.1.19 Furnish and install two new upgraded pumps of the same size, power rating, and compatibility with the existing system.

3.3.1.20 Ensure all new piping is restored to up-to-date insulation. The IMC specifies that pipe insulation must conform to IECC, be tested by ASTM E84 or UL 723, and have a maximum flame spread index of 25 and smoke-developed index.

3.3.1.21 Furnish and install four (4) new upgraded condenser water pumps (Pumps 1B & 2B; 75 HP, 230/460V, 1780 RPM) and (Pump 3B & 4B; 100 HP, 230/460V, 1780 RPM) or equivalent, fully compatible with existing piping and electrical infrastructure. Ensure Pump 1B & 2B are cross-connected so that it can operate on either dedicated chiller. Ensure that Pumps 3B and 4B are cross connected, so that it can operate on either dedicated chiller.

3.3.1.22 Demolish existing condenser water piping from pumps (1B, 2B, 3B, 4B) to the cooling towers and install new piping that is compatible in size, material, and function with the previous configuration. All new piping shall comply with applicable mechanical codes and site design standards

3.3.1.23 Furnish and install new piping that is compatible in size, material, and function with the previous configuration. All new piping shall comply with applicable mechanical codes and site design standards.

3.3.1.24 Furnish and install a new chemical feed system compatible with current water

treatment requirements. The system shall include all necessary tanks, dosing pumps, sensors, and controllers.

3.3.1.25 Furnish and install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

3.3.1.26 Have all new fire device installation work coordinated with the fire-alarm/fire-suppression service contractor

3.3.2 **HVAC Controls**

3.3.2.1 The government will provide the contractor with the Controls and Control Panel Specifications guideline for their use and reference.

3.3.2.2 Furnish and install all associated mechanical controls necessary for properly operating, controlling, alarming, and monitoring the HVAC systems and their related components. Control items to address shall include but are not limited to the following: VFD, smoke detectors, HVAC fan motor startup/ shutdown, and interface with existing fire detection system and fire alarm panel, condenser pumps, etc.

3.3.2.3 Furnish and install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

3.3.2.4 All devices shall utilize SCADA communication protocol and shall have manufacturer produced Protocol Implementation Conformity Statement (PICS) submitted and approved prior to equipment approval or release for purchase.

3.3.2.5 A building automation system schedule shall be created, and integrated to SCADA, for each necessary mechanical/ hvac equipment. The schedule shall use existing Holiday, 24/7 Operation Override, and Year End Shutdown (YES) Calendars.

3.3.2.6 Integrate HVAC controls into the existing Supervisory Control and Data Acquisition (SCADA) system.

3.3.2.7 Integration of HVAC controls shall be coordinated with the SCADA System Manager and shall be consistent with other existing HVAC units.

3.3.2.8 The equipment shall have BACnet TCP/IP based communication or contractor may utilize a supervisor/gateway/or BACnet router to bridge BACnet RS-485 network to the BEP IP network. The Contractor shall be responsible for properly terminating and shielding the RS-485 communication trunk.

3.3.2.9 Contractor to provide, label, terminate, test and certify any needed IT Network drops back to nearest enterprise network closet form the controller(s).

3.3.2.10 The existing SCADA is a group of servers and clients with primary, redundant, and Historian SQL servers, and multiple SCADA clients. All reside on the BEP isolated network. Below is the following configuration of the SCADA system

3.3.2.11 The SCADA system is running CITECT 8.X (2016) with redundancy failover.

3.3.2.12 All the graphics, menu tree structure, alarm points, and trend data reside at the server level and shall be updated for new equipment.

3.3.2.13 Historical trend data is stored on the Primary and Redundant SCADA Trend servers and is transferred to the Historian SQL server, where reports are generated.

3.3.2.14 The existing HVAC controls utilized in the building are Facility Explorer.

3.3.2.15 Field and panel wiring shall match the existing color coding used in existing HVAC control panels. All components and wiring shall be submitted for approval.

3.3.2.16 Contractor shall provide one spare controller of each type, as well and one spare room sensor.

3.3.2.17 Contractor must use controllers compatible with software existing at BEP. Otherwise, contractor must provide any and all software, licenses, cables to program/commission the controller.

3.3.2.18 Contractors shall use BEP laptops to do initial programming, checkout, and commissioning of the controllers at BEP.

3.3.2.19 No Contractor laptops are allowed to connect to any electronic devices at BEP. Including new HVAC controls. (IT security requirement).

3.3.2.20 Install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

3.3.2.21 All new fire device installation work coordinated with the fire-alarm/fire-suppression service contractor.

3.3.3 **New structural steel platform**

3.3.3.1 The Contractor shall submit approved shop drawings for the new structural steel platform for the new cooling towers, as per the submitted and reviewed set of drawings listed in Section 3.3.2.2.

3.3.3.2 The Contractor shall erect, install, and finish the new structural steel platform for the new cooling towers as per the approved shop drawings and the submitted and reviewed set of drawings and specifications listed and based on Section 3.3.2.2 requirements.

3.3.3.3 The Contractor shall submit all necessary testing reports from an independent third-party certified testing agency for the newly constructed structural steel platform to BEP for review and records.

3.3.3.4 The Contractor shall provide all the equipment and scaffolding systems with all means and methods required for this project; and shall be responsible and liable to comply with all safety codes, standards, and regulations, and protect pedestrians, surrounding properties, and trees. The Contractor shall also be responsible for securing the appropriate permits from Washington, D.C., DC DOT, and local AHJ as needed.

3.3.3.5 The Contractor shall submit a project schedule outlining all required activities and tasks indicated on this SOW after two weeks of the date of the NTP (Notice To Proceed) issuance.

3.3.3.6 Weekly progress report and construction meeting shall be required during the construction phase

3.3.3.7 The Contractor shall submit required drawings / submittals / samples / … etc. under this section within ten weeks after the date of the NTP (Notice To Proceed) issuance.

3.3.3.8 Construction period under this section is expected not to exceed 730 calendar days from date of the NTP issuance. Upon the completion of the construction, statement of substantial completion shall be required along with an as-built drawings for the new structural steel platform.

3.3.3.9 All required drawings shall be submitted in PDF and AutoCAD versions.

3.3.3.10 Coordinate with the equipment supplier and confirm exact locations of the

new cooling towers on the new platform and provide all necessary supports to secure this equipment.

3.3.3.11 The Contractor shall provide skilled and certified labors with sufficient experience to perform and execute all construction activities and tasks listed on this SOW.

3.3.3.12 The Contractor shall safely and securely remove and clean regularly the site from all construction debris that includes all hazardous materials. At the completion of this project, the contractor shall remove and dismantle any equipment, tools, scaffolds, signs, etc., and ensure the building exterior and surrounding areas are safe for public use.

3.3.3.13 The Contractor shall comply with the current editions of all applicable standard, practices, codes, and methods. In the event of conflict between codes and standards of the organizations, the more stringent regulations shall govern. These standards and codes shall include, but not limited to:

IBC – International Building Code
IMC – International Mechanical Code
AISC – American Institute for Steel Construction
ACI – American Concrete Institute
ASCE 7 – American Society for Civil Engineers
ICC – International Code Council
NFPA – National Fire Protection Association
NEC – National Electrical Code
OSHA – Occupational Safety and Health Administration Standards
Current edition of ANSI, UL, Consumer Product Safety Commission, and the manufacturer's recommendations for polycarbonate selection, installation and painting
AHJs (Authorities Having Jurisdictions) ordinances and guidelines

3.3.4 **Roofing Replacement**

3.4.4.1 The contractor shall conduct site survey, review available drawings and documents, and design the new roofing system. Recommendations for alternative roofing systems shall be addressed based on the current market materials and roofing systems. This roof shall serve the cooling towers, water lines, pipes, conduits. etc., and all related equipment and that are supported by the new steel frame.

3.4.4.2 The design shall be the result of assessments, investigation and analysis performed by the contractor and the contractor's subcontractors. The contractor shall have a licensed engineers designing the new roofing system. The contractor shall have sufficient experience in installing similar roofing systems.

3.4.4.3 The design comprises a set of independent work package designs that together specify how to fulfill the intent and purpose of this contract. The work packages will be identified and selected in a way that allows fast tracking without risk to other work packages or the overall design. Hereafter, the unqualified word, "design," refers to either the overall design or an independent work package design depending on context.

3.4.4.4 The contractor shall procure design services from an independent, accredited architectural/engineering firm with previous experience relevant specifically to the scope and requirements herein.

3.4.4.5 The design will provide the best solution to fulfill the scope, requirements, concepts, and intents described herein and in pre-proposal meetings and visits.

3.4.4.6 The contractor shall provide signed and sealed design calculations, construction drawings, and specification package by a Washington, D.C. registered design professional for approval. The contractor shall provide a preliminary schedule 1 week after contract is awarded and at the kickoff meeting.

3.4.4.7 The contractor shall develop the design through an iterative process, moving to implementation only when a particular design increment represents a complete and final design with the full concurrence of the BEP.

3.4.4.8 The iterations at each design level will follow these steps with the explicit goal of creating a design that aligns with the intent and expectation of this contract.

- Investigate, assess, evaluate customer's stated objectives, goals, and requirements to develop clear and specific requirements that fill gaps.

- Demonstrate that the design has incremented to the appropriate level of maturity by providing supporting documentation.

- Participate in meetings and discussions with the intention of ensuring that the current design increment aligns to the objectives, goals, and requirements of the customer. Adjust as necessary.

- Demonstrate the aligned design increment at the specified level.

- A design increment is complete only when it meets the following criteria:

- Contains a design narrative describing the current design increment in general terms. This may just be an expansion of the scope statement from the previous increment's way forward recommendation. The narrative will include an estimate of the percent completion this increment provides.
- Contains sketches or plans to the appropriate level.
- Contains an updated report of assumptions, risks, and issues, along with recommendations for managing such.
- Contains an updated path forward describing the remaining work necessary for a complete design along with a recommendation for the scope of the next design increment.

- Expect the review process for each submittal to consume up to 10 business days.

- No design increment shall consume more than 4 calendar weeks.

Contractor shall use the following waypoint guidelines to gage progress of a design:

- 50% or greater
  - Scope narrative
  - Draft design drawings
  - Draft specifications
  - Draft construction milestone schedule
  - Draft cost estimate

- Final
  - Complete drawings, including general sheets, general notes, and general trades details. Documents should be provided in both PDF form as well as Layered DWG CAD files where applicable.
  - Complete specifications, incorporating BEP provided general requirements.
  - Complete milestone schedule at 1 week granularity.
  - Detailed cost estimate.

The contractor shall implement the accepted final design.

Contractor shall use this SOW along with the associated specifications, drawings, attachments as the basis for all work and deliverables.

3.4 **Construction requirement.**

a. The contractor shall remove and dispose existing Chiller / Power Plant Wing Roof roofing layers and materials and replace it with a new hazardous free roofing system. No records are available pertaining to the drawings of the existing roofing system or build up. It is unknown what the specific materials

are.

b. Existing roof flashings may need to be removed, reworked, and replaced in order to accommodate the new roof. Evaluating existing cast iron drains is required. Rework / replace these flashings and drains as needed after BEP review and approval.

c. New roof shall be one finished surface.

d. The contractor shall coordinate and perform the flashing around the new steel stub-ups for the new steel platform.

e. Existing roof insulation is to be replaced according to proper code requirement. The R-value for the roofing system must be brought into compliance with current energy codes. This shall be part of the design and part of the roof slab membrane replacement.

f. The Contractor shall be responsible for complying with all requirements and submitting all deliverables identified in 75D-07.0-04 EHS Contractor Requirements.

g. Prior to the commencement of the replacement work, the contractor shall water-test the existing drainage system to ensure no blockage of the drain causing overflow and leakage. After completion of the work, the contractor shall perform flood-test. Should Chiller / Power Plant Wing Roof fail this test and deficiencies exist, the contractor must rework these deficiencies to ensure that the roof is watertight with no signs of water ponding.

h. The Contractor shall provide all the equipment and will be responsible for coordination with other trades, and all means and methods required for this project; and shall be responsible and liable to comply with all safety, codes, standards, regulations, and protecting pedestrians, surrounding properties, and trees. All items damaged by the contractor will be repaired/replaced at no additional cost to the Government. Repair/Replacement of damaged BEP property is subject to approval by the Contracting Officer.

   In the event of conflict between codes and standards of the organizations, the more stringent regulations shall govern. These standards and codes shall include but may not be limited to:

   - IBC - International Building Code
   - IMC - International Mechanical Code
   - NFPA - National Fire Protection Association
   - NEC - National Electrical Code
   - OSHA - Occupational Safety and Health Administration Standards

i. All tasks shall be performed and completed in accordance with the drawings and specifications related to this SOW scope. Contracting Officer must approve all work and tasks that is beyond this SOW.

j. Though all FAR clauses are enforced, the contractor should understand the impact of FAR 52.236-3 fully prior to offering an executable proposal.

This means that the initial proposal of the contractor's team will not modify the scope and intent of these requirements. Based on pre-proposal site investigations, the proposal may further refine the level of detail defining the work means, methods and deliverables necessary to successfully meet the scope and intent of these requirements, capturing all costs prior to execution of the contract.

k. Accepted subsequent proposals may modify the scope and requirements based on unforeseen conditions or the will of BEP to add scope. In such cases, the contractor shall provide all resources necessary to develop a comprehensive proposal for BEP review. Proposals will only be considered for acceptance when they include the following:

- A description of the reason for proposal. What is the problem/issue? When did it become known? What is the impact? Why must it be resolved as additional scope on this contract?

- A detailed analysis of the change to scope, requirements and schedules (fragments and time impact analyses), including alternatives.

- For each alternative, include costs broken down to the level detail BEP requires to make an informed decision.

- All source information (e.g., detailed subcontractor quotes).

3.5 **Quality Requirements.**

Testing and Inspection - The Contractor is responsible for testing and inspections. The Contractor will submit names of all testing services to the COR for approval prior to any testing. All inspections and tests must be performed by an independent, third part firm/company. The independent, third-party firm/company shall be appropriately registered in the District of Columbia. The Contractor shall engage only in inspections and testing services companies, including independent testing laboratories that are prequalified as complying with the American Council of Independent Laboratories "Recommended Requirements for Independent Laboratory Qualification" and that specialize in the types of inspections and tests to be performed.

3.5.1 The Contractor shall submit a Schedule of Tests and Inspections. The Schedule of Tests and Inspections shall be prepared in tabular form and include the following

I. Specification section Number and Title
II. Description of test and inspection
III. Identification of applicable standards
IV. Identification of test and inspection methods
V. Number of tests and inspections required.
VI. Entity responsible for performing tests and inspections.

3.5.2 All tests and inspections shall be documented by certified written reports and test results. The independent, third part individual firm and/or testing company shall distribute copies of all testing and inspection reports to the COR, and the Contractor. The Contractor shall make additional copies as required and distribute to all affected Sub-Contractors and suppliers.

3.6 **Adherence to the Scope of Work.** Although this SOW identifies specific elements of construction, it is the Contractor's responsibility to provide all elements which are incidental for the functioning of the work to be provided. It is not intended for other deficiencies not related to the work identified in this scope, to be corrected. The Contractor shall limit its efforts only to work identified in this SOW.

3.7 **Storing Materials.** The Contractor shall be responsible for receiving, storing, and securing of all materials, equipment, and other items to be used in accomplishing the work. The Contractor shall be responsible for all equipment and materials brought onto the BEP property and shall replace damaged and/or stolen items at its own expense.

3.8 **Incidental Cutting and Patching.** The Contractor shall be responsible for all cutting and patching incidental to the work described herein. The Contractor shall be responsible for identifying the location of existing utilities and for avoiding damage while completing this work. The Contractor shall promptly repair and/or replace any damage to structures, elements, utilities, etc., occurring due to construction, the cost which shall be borne by the Contractor.

3.9 **Existing Utility Interruptions.** The Contractor shall not interrupt any utilities serving the facilities occupied by the BEP unless permitted under the following conditions and then only after arranging to provide temporary utility services to maintain facility function.

- Notify the BEP COR, not less than **seven (7) calendar days** in advance of the proposed utility interruption.
- Do not proceed with utility interruptions without notification to the COR and/or the CO

3.10 **Pre-Construction Meeting.** A pre-construction meeting is required. The meeting will include coordination with the BEP, project procedures, staging, phasing, sequencing, lines of communications, project deliverables, roles and responsibilities, safety and quality requirements.

3.11 **Site Visits.** Site visits and construction observations are required once construction commences, in coordination with the BEP. The Contractor shall prepare an agenda for the weekly progress meetings, compile the meeting minutes and distribute to the team members and attendees within three (3) calendar days after the meeting. These weekly progress meetings shall include representatives from the Contractor and BEP team.

3.12 **Parking.** There is no on-site construction parking, but vehicles are permitted to drop-off materials upon request.

3.13 **Dust Control.** Dust control measures shall be implemented to minimize dust from traveling within the designated area.

3.14 Occupant Protection must be implemented within the BEP buildings by installing safety cones, rails or other barriers to prevent injury from unauthorized entry of personnel into the work area.

3.15 Upon completion of work, the Contractor shall provide training to the BEP Power Plant and Facilities Maintenance personnel.

3.16 Prior to commencing demolition or construction activities, the Contractor shall submit for review and approval for the following:

I. Preliminary Working Schedule-Due 2 weeks after the kickoff meeting.
II. Safety & Fire Protection Plan
III. Activity Hazard Analyses (AHAs)
IV. Crane Lift Plan
V. Fall Protection Plan
VI. Quality Control Plan
VII. Phasing/Staging Plan
VIII. Installers' Qualifications
IX. Hazardous Waste Plan (if applicable)

3.17 **Construction Documents Criteria.**

3.17.1 Action Submittals: The Contractor shall provide product data submittals for review and approval by the BEP before the purchase and installation of all new equipment.

3.17.2 As-Built Drawings are to be submitted fourteen (14) calendar days before requesting Substantial Completion walk-through. As-built drawings shall be in PDF and AutoCad 2021.

3.17.3 Construction Activities Arrange deliveries of products in accordance with construction schedules. Coordinate to avoid conflict with work and conditions at the site. Contractor shall minimize disruption to the daily operations of the BEP as much as possible.

3.17.4 Deliver products and materials in an undamaged condition, in the manufacturer's original container and/or packaging, with identifying labels intact and legible. Immediately on delivery inspect shipments to assure compliance with requirements of contract documents and approved submittals, and that products are properly protected and undamaged.

3.17.5 Construction deliveries will be permitted 0600 am to 1400 pm, Monday through Friday, provided parking and other BEP activities are not disrupted. When the COR determines the deliveries will be disruptive, deliveries must be scheduled during the weekend hours. All deliveries must be scheduled with the COR and/or CC within a minimum of 48 hours in advance of the delivery. Freight elevator is available to use.

3.17.6 Construction Work Schedule The Contractor shall submit a construction schedule to the CO and/or COR for approval. The schedule shall account for all divisions of work identified in the specifications and drawings. The schedule shall clearly show the completion date, duration and any milestones outlined in the specifications. The Contractor shall also ensure that the schedule is updated regularly or as specified by the COR.

3.17.7 Request for Outages At least three (3) weeks prior to the delivery of a large piece of equipment, and/or commencement of activities that will impact BEP's daily operations, such as utility outages, blocking or obstructing access through corridors, etc., the Contractor shall submit in writing to the CO and/or COR his/her schedule for this work. If this work will cause an interruption of necessary services, the work shall be conducted during off-hours, weekends, or holidays.

3.17.8 General The Contractor shall submit all Request for Information (RFI) in writing to the CO and/or COR and allow at least seven (7) calendar days for the COR to respond.

4.0 Supervision of Contractor's Employees Personnel assigned to render services under this contract shall always be employees of the Contractor and under the direction and control of the Contractor. Notwithstanding any other provisions of this contract, the Contractor shall always be responsible for the supervision of his/her employees in the performance of the services required hereunder.

4.1 During all time on the BEP premises, the Contractor's personnel shall comply with the rules and regulations governing conduct of personnel and operation of the facility.

4.2 If the Contractor plans to employ any non-English speaking personnel, he/she shall provide an on-site bilingual supervisor to serve as an interpreter.

4.3 H-18 Posting at Work Site. Upon commencement of work under this contract, the Contractor shall post a conspicuously located sign at the entrance of each work site. The following information is required to be posted:

1. Name of Prime Contractor

2. Name of Prime Contractor's On-Site Supervisor
3. Name of COR
4. Title of Work

4.4 Hazardous Material. The Contractor is responsible for coordinating and conducting all applicable lead-based paint (LBP) surveys and analyses for the work area(s) prior to demolition. The contractor shall forward all sample analysis results to the BEP LBP Program manager for review at least two weeks prior to the commencement of work. Any LBP abatement required to complete the project shall be conducted/coordinated by the Contractor. In addition, the Contractor shall submit a LBP abatement plan to the BEP LBP Program manager for review and approval two weeks after the Notice to Proceed is given for the commencement of work. The Notice to Proceed ONLY allows the contractor to begin coordinating logistics (i.e., badging, security clearances, etc.). It does not indicate that the abatement can commence. The LBP abatement plan MUST be approved, in writing, by the BEP LBP Program manager prior to the start of the abatement work. The plan MUST be approved, in writing, by the BEP LBP Program manager prior to the start of work.

4.5 Acceptance of Work

4.5.1 Substantial Completion Substantial completion of the work is defined herein as the point at which the work is complete in all respects except for a few minor items which are to be listed on the Contractor's punch list. Substantial Completion is inclusive of all accepted systems being fully operational. The Contractor shall demonstrate that all newly installed systems are fully operational. With the issuance of a substantial completion certificate, executed by the Contracting Officer (CO) and the Contractor, the BEP will take possession of the work and operate the installed systems/equipment. Prior to the issuance of an executed substantial completion certificate the Contractor shall have had all tests completed, witnessed, and approved by the BEP's authorized representative. The provision of all required training, operating and maintenance manuals, and other such requirements in this SOW shall be a prerequisite to the Contractor's notification requesting a substantial completion walk-through.

4.5.2 The Contractor shall notify the CO and/or COR in writing at least ten (10) calendar days prior to the estimated date of substantial completion and request a substantial completion walk-thru. The letter shall include a dated punch list as developed by the Contractor. The purpose of the walk-through is to review the Contractor's list for accuracy and to identify any additional items needing completion prior to final acceptance. The Contractor, shall, upon receipt of a partial or substantial completion punch list, correct his or her work as required within seven (7) calendar days. The substantial completion walk-through shall be performed by the BEP COR and/or CC.

4.5.3 The date of the BEP's acknowledgement of substantial completion shall establish the

Contractor's one year guarantee of workmanship and the manufacturer's product warranties. In the instance of a Partial Substantial Completion Certificate, the previous sentence shall apply to the work identified in the partial substantial completion certificate only. The Contractor shall ensure all warranty requirements, including manufacturers' warranties, in an electronic spreadsheet format that will be provided to the CO and COR.

4.5.4 A 30- year warranty, and a 6-year weather-tight warranty shall be provided for this roof project. Also, yearly inspection, for six years after the construction completion date and handing over this new roof project shall be included in this scope. Twice a year roof inspection is required for the first year, and once a year inspection is required thereafter for the following five year. The contractor shall submit an exhaustive report of the findings following each inspection. The Contractor shall ensure all warranty requirements, including manufacturers' warranties, in an electronic spreadsheet format that will be provided to the CO and COR

4.5.5 Prior to requesting Substantial Completion. The Contractor is required to submit

4.5.5.1.1 Warranties
4.5.5.1.2 O&M Manuals
4.5.5.1.3 Records of Waste Disposal (if applicable)
4.5.5.1.4 Appropriate Testing and Inspection Results
4.5.5.1.5 Final Punch List with Request for Certification that the Project is Substantially Complete
4.5.5.1.6 As-Built Drawings

4.6 Final Completion. The Contractor shall inform the CO and/or COR in writing at least seven (7) calendar days prior to the estimated date of the completion of the work and request a final acceptance inspection.

4.7 Deliverables. The Contractor tasks requirements summarizes the performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of work required for each requirement. These thresholds are critical to mission success.

#	Deliverable	Due Date	Recipient	Deliverable Format
1	Weekly report of work	At weekly construction meeting.	COR & Project coordinator	Word
2	Project schedule	Two weeks after NTP date.	COR	MS Project
3	35% Concept / Schematic Design	Two weeks after the project schedule.	COR	AutoCAD, PDF, 3 hard copies
4	65% Intermediate Design / Design Development	Two weeks after Government's response.	COR	AutoCAD, PDF, 3 hard copies

5	95% Pre-Final / Construction Documents	Two weeks after Government's response.	COR	AutoCAD, PDF, 3 hard copies
6	100% Final Design / Approved Documents	Two weeks after Government's response.	COR	AutoCAD, PDF, 3 hard copies
7	Issue for Construction (IFC) Final signed and sealed drawings, calculations, and specifications submission.	Two weeks after Government's response.	COR	AutoCAD, PDF, 3 hard copies
8	Submit all products, equipment, material, and diagrams for approval prior to installation.	Two weeks after IFC stamp.	COR	PDF / Samples
9	Safety & Fire Prevention Plan, Environmental Health and Safety Plan, Safety Data Sheets, Waste Management Plan, Crane Lift Plan and Lock Out-Tag Out Plan, Hazardous Plan, Quality Control Plan and Phasing/Staging Plan.	Three weeks after NTP.	COR	PDF
10	Statement of Substantial Completion	180 Calendar Days after the NTP.	COR	PDF
11	Draft As-built drawings	(14) days prior to the Substantial Completion walk-through.	COR	AutoCAD, PDF
12	Operation & Maintenance (O&M) manuals and Final Approved As-built drawings	Two weeks after project substantial completion.	COR	AutoCAD, PDF
13	Project Meeting	Weekly.	COR	MS Word/PDF
14	Project Meeting Minutes	Weekly.	COR	MS Word/PDF
15	Daily Construction Reports	Weekly.	COR	MS Word/PDF
16	Field Condition Reports	Biweekly .	COR	MS Word/PDF

4.8 Performance/ Delivery Period.

The period of performance for this contract shall take no more than **730 calendar days** from the issuance of the Notice to Proceed (NTP) to completion.

4.9 Place/ Location of Performance/ Delivery.

4.9.1 Hours of Operation The contractor is responsible for conducting business, between the hours of , Monday-Friday, between 0600 am and 1600 pm (EST) Monday thru Friday except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. Hours may change according to the agency needs. Weekend work is not required but may be necessary depending on the areas in question. Demo can take place during normal hours. Demo is permitted during normal business hours. The contractor shall coordinate with the government regarding hour of operation for BEP Year End Shut Down, which are generally the days between Christmas and New Years Day. For other than firm fixed price contracts, the contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this SOW when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

Federal Holidays

New Years Day	1st day of January
Martin Luther King Jr.'s Birthday	3rd Monday of January
Presidents Day	3rd Monday of February
Memorial Day	Last Monday of May
Juneteenth	19th day of June
Independence Day	4th day of July
Labor Day	1st Monday of September
Columbus Day	2nd Monday of October
Veterans Day	11th day of November
Thanksgiving Day	4th Thursday of November
Christmas Day	25th day of December

4.9.2 Place of Performance The work to be performed under this contract will be performed at BEP's DCF 14th & C Streets, S.W., Washington, D.C.

4.9.3 Special Requirements.

I. Security Requirements:The Government, for the purpose of site access control, operates an entry control point. Contractors, subcontractors, and suppliers will be required to show a valid government issued photo identification to enter the site or will be denied entrance until clearance is granted by the Contracting Officer. The Government's operation of an entry control point does not relieve the Contractor of its responsibility for the security and control of its own equipment, material, and personnel.

All Contractor personnel must stop at the Entry Control Point (Processing Center) to receive an "Escorted Visitor Badge" which must be worn at all

times while on BEP premise. Escorts will be provided by the BEP personnel.

Any Contractor personnel who will be on site for 30 aggregate days or less in a calendar year will be required to be escorted and receive a Visitor Escort badge.

Any contractor personnel who require access to the site for more than 30 aggregate days in a calendar year, BEP form 2276, Contractor Access Request Form (CAR), must be completed. This form is submitted to Personnel Security to request a background investigation on those Contractors that require short term or long term access to the facility. The prime Contractor will furnish the names of those contractors meeting this requirement to the COR in advance of any work being started on the contract allowing the Security Office to conduct said checks. No access will be granted to perform work on the contract until they have been cleared by Personnel Security. In addition, computer based training will be required to receive a badge. The contractor will be notified by the Contracting Officer's Representative (COR) when clearance has been granted.

Any Contractor personnel who require Short Term access which is access on site for more than 30 aggregate days will be required to complete a Criminal History Check release form and have an FBI fingerprint check completed.

The contractor must provide proof of citizenship by providing an original or certified copy of a state issued birth certificate or valid US Passport; or proof of legal residency and employment authorization; an original social security card; and proof of identity such as a valid driver's license or state identification card.

The release forms will be provided to the contractor during their fingerprint appointment with Personnel Security at the site. These contractors will be issued a Visitor Escort badge.

Any Contractor personnel who will require Long Term access which is access on site for more than 180 aggregate days in a calendar year or the contract is for 180 days or more, will be required to undergo at a minimum a **(Tier 1)** background investigation in addition to the requirements for Short Term access, prior to the beginning of that individual's work on site. Personnel Security will determine the level or tier at which the background investigation will be completed.

The Contractor will furnish the COR a listing of each employee meeting this requirement. All contractor personnel that require this type of background investigation will submit an eApplication through use of the Office of Personnel Management's electronic Questionnaire for Investigative Processing (e-QIP). A point of contact from the company will be required for Personnel Security to furnish information regarding completion of the eApplication and required release forms. Contractor personnel will have 10 calendar days to complete the eApplication or access to the facility may be denied. If the contractor fails to complete the eApplication in the allotted time, their profile will be deleted and the COR will be notified. The COR may resubmit the request using a new CAR form. If the contractor fails to complete the eApplication on the second request, no further requests can be submitted and the contractor will not be allowed access to the facility to include access as a Visitor.

Contractors that require a background investigation must submit to the Personal Identity Verification (PIV) process and be "identity proofed". Contractors will be required to provide two valid and unexpired authorized forms of identification with at least one being issued by a federal or state government and contain a photograph. Those Contractor personnel that undergo a background investigation for long term access will receive a badge with a photograph and will not be required to be escorted in areas where they have badge access.

BEP has the right to deny access to the facility to anyone who is determined to be unsuitable for access to the facility. The determination of unsuitable does not reflect on the individual's suitability for employment with the contract company, only access to the facility. All operators of motor vehicles shall have a valid and unexpired Real ID-Compliant State Driver's License while operating a motor vehicle on the premises of the District Currency Facility.

All packages carried by persons entering or leaving the Bureau must be opened for inspection by the uniformed Police. Personal packages may not be brought into the Bureau (except lunches, work clothing, and equipment necessary for work). Unauthorized removal of Government property is punishable by law.

All Contractor personnel (except when issued a picture badge) must be escorted at all times while on Bureau premises.

Contractor personnel issued this badge may proceed without Bureau employee escort in corridors, warehouse areas, etc., except security areas where entry is controlled by a police check point, or other areas where securities are processed. Personnel issued these badges may escort visitors authorized by the COR in areas where they have badge access.

II. Physical Security The Contractor shall be responsible for safeguarding all government equipment, information, and property provided for Contractor use under this agreement. This agreement requires access into the Controlled Industrial Area of the BEP. The Contractor personnel shall recognize that BEP deals with sensitive information and other proprietary information, which are valuable, special, and unique assets. All information pertinent to BEP's information technology resources is highly sensitive and may not be released to third parties without the express, written consent of BEP. This information requires strict control procedures to preclude unauthorized disclosure or dissemination. Information pertinent to personnel is protected from unauthorized disclosure under the provisions of the Privacy Act of 1974 (5 U.S.C. 552a). Contractor personnel shall not divulge, disclose, or communicate in any manner, any information to any third party without prior written consent of the BEP. Contractor personnel shall protect information and treat it as strictly confidential and will abide by the security requirements of BEP. A violation of this paragraph shall be a material violation of this agreement. Should the Contractor receive or suspect they have learned of sensitive or classified information, the Contractor shall identify and report suspicions to the COR.

III. Key Control The Contractor shall establish and implement methods of making sure all keys/key cards issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. **NOTE All references to keys include key cards.** No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop procedures covering key control that shall be included in the Quality Control Plan. Such procedures shall include the turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any occurrences of lost or duplicate keys/key cards to the Contracting Officer.

In the event keys, other than master keys, are lost or duplicated, the Contractor shall, upon the direction of the Contracting Officer, re-key or replace the affected lock or locks; however, the Government, at its option, may replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the lock or locks shall be

deducted from the monthly payment due the Contractor. In the event a master key is lost or duplicated, all locks and keys for that system shall be replaced by the Government and the total cost deducted from the monthly payment due to the Contractor.

The Contractor shall prohibit the use of government-issued keys/key cards by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of assigned work in those areas, or personnel authorized entrance by the Contracting Officer.

IV. Special Qualifications The Contractor and all Sub-Contractors shall have extensive experience with installing HVAC systems for industrial and federal buildings. The experience shall specialize in mechanical contracting with an emphasis on large HVAC installations for existing buildings. Contractor or sub-contractor performing integration to SCADA shall be certified as Citect SCADA Certified Expert (CSCE). Johnson Controls International (JCI)/Niagara training and AVEVA Advanced System Integrator (or ASI Controls) certifications for SCADA are required. Contractor or sub-contractor performing controls installation shall be an FX ASI (Facility Explorer Authorized System Installer.) The contractor shall be responsible for ensuring all employees possess all required licenses and training for operating equipment and vehicles used in the execution of this contract. Use of any chemicals shall meet EPA guidelines, scafederal regulations, and District of Columbia (DC) Department of Health requirements.

V. Post-Award Conference/Periodic Progress Meetings. The Contractor agrees to attend any post-award conference convened by the contracting activity or contract administration office in accordance with Federal Acquisition Regulation Subpart 42.5, Post Award Orientation. The Contracting Officer, COR, and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the Contracting Officer will apprise the contractor of how the government views the contractor's performance and the contractor will apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

VI. Contracting Officer Representative (COR).

a. The COR will be identified by a separate letter by the CO. The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following

functions assure that the Contractor performs the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, and specifications; monitor Contractor's performance and notify both the Contracting Officer and Contractor of any deficiencies; coordinate availability of government furnished property; and provide site entry of Contractor personnel. A letter of designation issued to the COR, a copy of which is sent to the Contractor, states the responsibilities and limitations of the COR, especially with regard to changes in cost or price, estimates, or changes in delivery dates. The COR is not authorized to change any of the terms and conditions of the resulting order.

b. Key Personnel. The following personnel are considered key personnel by the government specialized technical personnel, project manager and superintendent. The contractor shall provide a contract manager who shall be responsible for the performance of the work. The name of this person and an alternate who shall act for the contractor when the manager is absent shall be designated in writing to the contracting officer. The contract manager or alternate shall have full authority to act for the contractor on all contract matters relating to daily operation of this contract. The contract manager or alternate shall be available between 0600 am to 1600 pm, Monday thru Friday except Federal holidays or when the government facility is closed for administrative reasons.

Resumes of Key Personnel shall show demonstrated experience in the position they're being considered for. The continued full-time availability of Key Personnel shall be formally confirmed in writing to the Government at time of award. The following personnel are considered "Key" Personnel.

VII. Specialized Technical Personnel (or equivalent labor category) Specialized Technical Personnel shall have a bachelor's degree from a U.S. accredited university or college with a major in engineering, architecture, or related technical field. Personnel shall demonstrate work experience, licenses (Professional Engineering) and technical competence of electrical, mechanical, structural/civil, architectural, fire protection, or environmental requirements covering the last five (5) years.

VIII. Project Manager (or equivalent labor category) The Project Manager shall have a bachelor's degree from a U.S. accredited university or college with a major in engineering, architecture, or related technical field, and/or at

least five years' experience in managing architectural, engineering or construction projects of similar size and scope and five (5) years' experience in Historical Government Buildings. The Project Manager shall be employed by the Prime Contractor.

IX. Project Superintendent (or equivalent labor category) The Project Superintendent shall have a minimum of five (5) years' experience as a superintendent on projects similar in size and scope, and five (5) years' experience in Historical Government Buildings and be employed by the Prime Contractor. The project superintendent must be on-site at all times and have overall responsibility for all operations on the jobsite.

X. Key Personnel Substitution Process The Government reserves the right to request replacement of any personnel who are deemed incompetent, careless, unsuitable or otherwise objectionable, or whose continued use under this contract is deemed contrary to the best interests of the Government. The COR will give notice, in writing, to the Contractor Point of Contact (POC) of such need for substitution. Within 48 hours of receipt of notification, the Contractor POC shall provide personnel replacement resumes to the COR. The COR will notify the Contractor POC, in writing, if the Contractor provided replacement personnel is approved. The COR will notify the Contracting Officer (CO) of the approval, and CO will issue a modification to formally replace the personnel.

During the performance of this contract, the contractor shall make no substitutions of key personnel unless the substitution is necessitated by illness, death, or termination of employment. The contractor shall notify the COR of this contract no later than five (5) calendar days after the occurrence of any of these events. The contractor shall include the circumstances necessitating the proposed replacement of the key personnel and shall provide to the Contracting Officer the name, resume of the proposed replacement. All replacement personnel require written approval from the Contracting Officer prior to their beginning work under this contract. When required, the contractor shall provide replacement employees that comply with Sections 4.9.3.X **Key Personnel Substitution Process** of this SOW. Any disputes regarding the substitution of key personnel after resume review will be decided by the Contracting Officer. The Government reserves the right to require the contractor to reassign key contractor employees who are deemed incompetent, careless, unsuitable, or otherwise objectionable, or whose continued use under this contract is deemed contrary to the best interests of the Government. The Contracting Officer will provide a written notice to the contractor of such reassignment.

XI. Identification of Contractor Employees All contract personnel (to include subcontractors) attending meetings, answering Government telephones, and working in other situations where their contractor status is not obvious

to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. They must also ensure that all documents or reports produced by Contractors are suitably marked as Contractor products or that Contractor participation is appropriately disclosed. All Contractor personnel shall clearly identify themselves as a Contractor and shall be required to wear identification (ID) badges in a visible location on the front of exterior clothing, above the waist, (except when safety or health reasons prevent such placement). ID badges shall be worn at all times when performing work. ID badges shall have an identification picture and shall state the name of the individual and the company they represent. A listing of issued identification cards shall be furnished to the Contracting Officer prior to the contract performance date and updated as needed to reflect Contractor and Subcontractor personnel changes. Contractor personnel are prohibited from possessing weapons, firearms or ammunition, on themselves or within their Contractor-owned or privately owned vehicle while on BEP premises.

XII. Supervision of Contractor Employees. The Government will not exercise any supervision or control over Contractor or subcontractor employees while performing work under the contract. Such employees shall be accountable solely to the Contractor, not the Government. The Contractor, in turn, shall be accountable to the Government for Contractor or subcontractor employees.

XIII. Data Rights. The Government has unlimited rights to all documents/material produced under this contract. All documents and materials, to include the source codes of any software, produced under this contract shall be Government owned and are the property of the Government with all rights and privileges of ownership/copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the Contracting Officer. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. This right does not abrogate any other Government rights.

XIV. Organizational Conflict of Interest. Contractor and subcontractor personnel performing work under this contract may receive, have access to or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications or work statements, etc.) or perform evaluation services which may create a current or subsequent Organizational Conflict of Interests (OCI) as defined in FAR Subpart 9.5. The Contractor shall

notify the Contracting Officer immediately whenever it becomes aware that such access or participation may result in any actual or potential OCI and shall promptly submit a plan to the Contracting Officer to avoid or mitigate any such OCI. The Contractor's mitigation plan will be determined to be acceptable solely at the discretion of the Contracting Officer and in the event the Contracting Officer unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the Contracting Officer may affect other remedies as he or she deems necessary, including prohibiting the Contractor from participation in subsequent contracted requirements which may be affected by the OCI.

XV. Non-Personal Service statement The Contractor employees performing services under this Contract will be controlled, directed, and supervised at all times by management personnel of the Contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in Contract. Contractor employees will perform their duties independent of, and without the supervision of, any Government official or other Government Contractor. The tasks, duties, and responsibilities set forth in this Contract may not be interpreted or implemented in any manner that results in any Contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions set forth in Parts 7.5 and 37.1 of the FAR.

XVI. **Applicable Documents**

Attachment 9 75D-04.0-08 Environmental, Health and Safety (EHS) Form Construction Safety Permit (After Award)
Attachment 10 75D-07.0-04 EMS Document EHS Contractor Requirements (mandatory compliance)
Attachment 11 75D-04.0-11 Fall Protection Procedure (mandatory compliance)
Attachment 12 75D-04.0-01 Emergency Action and Fire Prevention Plan (mandatory compliance)
Attachment 13 Mutual Multi Party Non-Disclosure Agreement (NDA) (After Award)
Attachment 14 Evapco performance data.pdf
Attachment 15 M02-schematic-BEPVFD Model Chiller Planty Flow Schematic mechanical 2.pdf
Attachment 16 M03-vfd plan-BEPVFD.pdf
Attachment 17 M04-Cooling Tower piping plan-BEPVFD M-4.pdf
Attachment 18 Plate & Frame HX Information.pdf
Attachment 19 Structural Support for VFDs.pdf
Attachment 20 Tower_submittal.pdf
Attachment 21 BEP-E1R613 Model Chiller Plant Electrical.pdf

Attachment 22 BEP-E2R0613 Layout1 Chiller Plant Electrical 2.pdf
Attachment 23 BEP-M1R613 w chiller Layout1.pdf
Attachment 24 Cooling Tower Information BEP Chilled Water Plant.pdf
Attachment 25 ax-dcm-06 Main Roof Top.pdf
Attachment 26 M-DCA-09-1001023-HV.pdf
Attachment 27 ax-dca-09 Annex Roof Top.pdf
Attachment 28 Chiller Plan Upgrade M-1_120508 Model (1).pdf
Attachment 29 Chiller Plan Upgrade M-1_recover.pdf
Attachment 30 Pictures
Attachment 31 Pictures
Attachment 32 Pictures

5.0 Requirements for Handling Sensitive, Classified, and/or Proprietary Information

5.1 Complete appropriate NDA's per DTAP policy as required.

6.0 Government Furnished Resources and Information

6.1 GOVERNMENT FURNISHED ITEMS AND SERVICES

All property equipment and services shall be provided by the contractor. If the contractor requires access for dropping off or picking up materials for specific jobs, entrance to the loading dock can be granted with advanced notice to the COR. No parking shall be provided by BEP for other purposes.

7.0 CONTRACTOR FURNISHED ITEMS AND RESPONSABILITIES

The contractor shall supply all equipment required to perform the specific tasks listed in Section III. Any waste created by the contractor shall be disposed of according to local and federal regulations. Supplies used shall be of the type, composition, quality, and rate standard in the professional industry for industrial hygiene services. All supplies or chemicals brought onsite shall be reported to and approved by the COR or his representative prior to use. Labels and Safety data sheet (SDS) sheets shall also be provided. X-ray diffraction (XRF) and other industrial hygiene equipment shall be calibrated and maintained according to federal requirements and manufacturer recommendation.

END OF DOCUMENT